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Safety and efficacy of inpatient pulmonary rehabilitation for patients hospitalized with an acute exacerbation of chronic obstructive pulmonary disease: a systematic review protocol

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Keywords:	Respiratory infections < THORACIC MEDICINE, Chronic airways disease < THORACIC MEDICINE, REHABILITATION MEDICINE

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Safety and efficacy of inpatient pulmonary rehabilitation for patients hospitalized with an acute exacerbation of chronic obstructive pulmonary disease: a systematic review protocol

Authors:

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ABSTRACT

Introduction: Pulmonary rehabilitation (PR) following an acute exacerbation of COPD (AECOPD) reduces the risk of hospital admissions, and improves physical function and health-related quality of life. However, the safety and efficacy of in-hospital PR during the most acute phase of an AECOPD is not well-established. This paper describes the protocol for a systematic review with meta-analysis to determine the safety and efficacy of inpatient acute care pulmonary rehabilitation during the hospitalization phase.

Methods and analysis: Medical literature databases and registries MEDLINE, EMBASE, PEDro, CINAHL, CADTH, CENTRAL, AMED, WHO trials portal, and ClinicalTrials.gov will be searched for articles from inception to June 2020 using a prespecified search strategy. We will identify randomized controlled trials that have a comparison of in-hospital PR with usual care. PR programs had to commence during the hospitalization and include a minimum of two sessions. Title and abstract followed by full-text screening will be conducted independently by two reviewers. A meta-analysis will be performed if there is sufficient homogeneity across selected studies or groups of studies. The Population, Intervention, Comparator, Outcomes, and Study characteristics framework will be used to standardize the data collection process. The quality of the cumulative evidence will be assessed using the Grading of Recommendations, Assessment, Development and Evaluations framework.

Ethics and dissemination: AECOPD results in physical limitations which are amenable to PR. This review will assess the safety and efficacy of in-hospital PR for AECOPD. The results will be presented in a peer-reviewed publication and at research conferences. No ethical review is required for this study.

Prospero registration number: registration submitted

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Keywords: Respiratory infections, Chronic airways disease, Rehabilitation Medicine

Strengths and limitations of this study

- This will be the first study to evaluate the safety and efficacy of pulmonary rehabilitation (PR) specifically for the in-hospital period for patients admitted for an acute exacerbation of chronic obstructive pulmonary disease.
- This review benefits from a comprehensive search strategy across bibliographic databases, trial registries, scientific conferences, and a grey literature database, ensuring the inclusion of as many relevant articles as possible.
- There may be studies that will be excluded that examine efficacy of inpatient pulmonary rehabilitation but do not have a discharge measurement, which is required for this analysis.

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INTRODUCTION

Frequent acute exacerbations are a common feature of chronic obstructive pulmonary disease (COPD), especially for patients with moderate to severe disease.[1] Although most acute exacerbations of COPD (AECOPD) can be managed in the outpatient setting, many patients require hospital admission. The cost of these hospitalizations is immense, representing more than 25% of the total health care cost of COPD in the United States.[2] Decreases in quality of life (QoL), lung function, quadriceps muscle force and physical activity have been reported in patients during and after hospitalization for AECOPD which, in many cases, are permanent changes.[3] Pitta et al,[4] reported that hospitalized patients with COPD show little improvement in their activity levels during their hospital stay, and by discharge only 9% of their day was spent standing or walking. To counteract these changes, pulmonary rehabilitation (PR) during or soon after an AECOPD is recommended.[5]

Most research on PR has focused on the stable COPD patient attending an outpatient program. However, a 2016 Cochrane systematic review by Puhan et al,[5] examined the effects of "early" PR (ie during or shortly after an AECOPD) on future hospital admissions and other patient-related outcomes such as mortality, health-related QoL, and exercise capacity in patients with COPD. The authors reported that early inpatient PR reduced the risk for hospital admissions, decreased mortality, and improved health-related QoL. There was also a marked improvement in exercise capacity. However, the Puhan et al. review,[5] included studies in which the PR intervention began in the hospital and, in some cases, continued after discharge. The measures of effect included the post-discharge period, when the patients were recovering at home. It is possible that much of the benefit of PR during an AECOPD is seen during this post-discharge period. What is not clear is if a PR intervention delivered during the hospital admission

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is safe and results in changes in health status and physical function prior to discharge. There has been some controversy regarding the safety of rehabilitation during this very acute phase, when patients are most ill with an AECOPD.[6] Clinicians who are considering implementing in-hospital PR for patients soon after admission must be confident about the efficacy and safety of these programs for their patients. Therefore, the purpose of this systematic review is to evaluate the safety and efficacy of in-hospital PR for patients admitted to hospital for an AECOPD.

METHODS AND ANALYSIS

This protocol has been prepared using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) guidelines.[7]

Types of Studies

We will select randomized controlled trials (RCTs) that have a comparison of in-hospital PR with usual care. We will include full-text, conference abstracts, and unpublished data in any language.

Types of Participants

This review will include studies of patients aged 19 years or older with a clinical diagnosis of COPD. The patient must be capable of physical activity (some form of movement whether it be active or stationary) and are hospitalized for an AECOPD at the time of the study. Any patients in an outpatient setting or in-patients admitted to hospital for any other reason besides their AECOPD are excluded from this study.

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The intervention of interest includes any rehabilitation program that involves mobilization, exercise, or ambulation started while the patient is still hospitalized for an AECOPD. The

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Outcome

We will include studies that report measurable pre- and post-intervention data prior to discharge for one or more of the outcomes of interest. The primary outcomes are health-related QoL, functional exercise capacity, and adverse events. The secondary outcomes are hospital readmissions within 30 days, and hospital length of stay (LoS).

Information sources

Bibliographic databases MEDLINE, Embase, the Cumulative Index to Nursing and Allied Health Literature (CINAHL), the Physiotherapy Evidence Database (PED*ro*), and trial registries (Cochrane Controlled Register of Trials (CENTRAL), the US National Institutes of Health Ongoing Trails Register (ClinicalTrials.gov), the Allied and Complementary Medicine Database (AMED), PsycInfo and the World Health Organization (WHO) trials portal) will be searched from their inception to June 2020 for potentially relevant articles using pre-defined search strategies (Appendix 1). Other searches will include the Canadian Agency for Drugs and Technologies in Health (CADTH) and hand-searches of meeting abstracts from the American Thoracic Society and the European Respiratory Society scientific conferences. There will be no

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restriction on the language of the publication. A manual search of the reference lists of all included studies will be conducted to check for other possibly relevant articles.

Search strategy

This review will utilize both keyterm search terms as well as MeSH (Medical Subject Headings) terms in both MEDLINE and CENTRAL. In Embase we will utilize Emtree terms and in CINAHL we will utilize CINAHL headings. A key "term" search strategy will be employed for the PED*ro*, ClinicalTrials.gov, and WHO trials portal database. The proposed search strategies for MEDLINE is listed in Appendix 1 and will be adapted for use in the other databases. An academic librarian is a member of the study team and is involved in all aspects of developing the search strategies. The reference lists of all primary studies and review articles will be checked for additional references.

Selection of studies

The selection of studies for inclusion will be conducted in two stages. Two reviewers will independently screen the titles and abstracts of all potential studies identified in the search. The studies will be coded as "Yes" or "No". The full-text publications of included citations will then be screened by two reviewers according to our inclusion and exclusion criteria. Studies not in English will be translated using Google Translate. Any discrepancies at any stage will be resolved by discussion and/or consultation with a third reviewer.

Data collection process

Two reviewers will independently collect information from all selected studies using a pre-specified data extraction form. The form will be piloted on the first six selected studies and refined, as necessary. Any discrepancies will be resolved by discussion and/or consultation with a third reviewer. We will contact authors of eligible studies for any missing data. Using the

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Population, Intervention, Comparator, Outcomes, Study Characteristics framework (PICOS) we will collect information on about study design, interventions, participants, outcome measures and adverse events to systematize our data extraction (Table 1).

Table 1. PICOS data elements

Population	• sample size • age • gender • severity of condition • baseline lung function • withdrawals
Intervention	• intervention • timing • number of sessions • frequency of intervention • supervision provided
Comparators	• identification and definition of unexposed individuals
Outcomes	• preintervention and postintervention values of primary and secondary outcomes • time points of when data was collected
Study characteristics	• study design • date of study

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	• funding for studies and notable conflicts of interested of trial authors

Risk of bias in individual studies

Risk of bias assessment will be conducted independently by two reviewers on all studies using the criteria outlined in the Revised Cochrane Risk of Bias Tool for Randomized Trials.[8] Any disagreements will be resolved by discussion and/or consultation with a third reviewer.

Data analysis

To assess differences in health-related QoL, we will determine whether estimates and 95% confidence limits between study groups exceed the minimal important difference (MID) for the Chronic Respiratory Questionnaire (± 0.5 points),[9] the St George's Respiratory Questionnaire (± 4 points),[10] the COPD Assessment Test (2 points),[11] the 6-minute walk test (≥ 30 meters),[12] and the incremental shuttle walk test (≥ 47.5 meters).[13] The unit of analysis was the participant. Patients who receive the usual care will be used as the control group in this study. Data will be synthesized by calculating mean differences and pooled odds ratios using random-effects models in Review Manager 10.[14]

Heterogeneity will be assessed using forest plots and the I^2 statistic. Heterogeneity will be examined using subgroup analysis and by analyzing items from the quality assessment that indicate high risk versus low risk of bias (for example, blinding during data collection procedures). If we find the included studies are sufficiently homogenous in terms of design, study population and outcomes, we will conduct a meta-analysis to calculate a pooled effect estimate. If prohibited by heterogeneity, a narrative synthesis will be conducted instead.

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In order to account for selective reporting within studies, we will assess all available protocols for the included studies using the trial registries ClinicalTrials.gov and the WHO trials portal. Publication bias across studies will be assessed using funnel plots.

The quality of the cumulative evidence for each outcome will be assessed using the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) framework which includes the risk of bias, consistency, directness, precision and publication bias.

Patient and Public Involvement

There was no patient involved in the development of this study.

ETHICS AND DISSEMINATION

Ethical review is not required as this study is a systematic review. It is our intention to submit the results of our review for peer-reviewed publication and to present our findings at scientific conferences.

DISCUSSION

Pulmonary rehabilitation plays an important role in the management of AECOPD. A large body of evidence shows that pulmonary rehabilitation in patients with AECOPD helps to improve QoL, symptom management, and disease prognosis;[15] however, to our knowledge, there has been no review to assess the acute in-hospital use of pulmonary rehabilitation for AECOPD. This review will determine the safety and efficacy of initiating physical rehabilitation, in patients with AECOPD, during their hospitalization and will provide suggestions for future guidelines and recommendations.

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Contributors

KZ, JG, and PC drafted the protocol and developed the study design and the data extraction form. AK, JC, AE, SC, HD, HP, and PC contributed to the development of the search strategies. All authors read and approved the final version of the manuscript for submission.

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Competing interests

None declared.

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Appendix 1. Database search strategies

Database: MEDLINE® ALL <Inception to June, 2020>

Platform: Ovid SP

Search Line	Search Item
1	exp pulmonary disease, chronic obstructive/
2	exp lung diseases, obstructive/
3	(COPD or COAD or AECOPD or COBD or AECB or (acute exacerbation adj2 (COPD or chronic obstructive pulmonary disease))).mp
4	((chronic or obstruct*) adj3 (airflow or airway* or bronch* or lung* or pulmonary or respirat*)).mp
5	(emphysema or lung hyperinflation or hyperlucent lung or enlarged alveolar spaces).mp
6	1 or 2 or 3 or 4 or 5
7	exp hospitalization/
8	exp inpatients/
9	(((transitional care or step down or subacute or acute care or medical acute or critical care) adj3 (center* or centre* or unit* or floor* or ward* or clinic*)) or (hospital* or intensive care or ICU or inpatient* or in-patient* or patient* or emergency department*)).mp
10	7 or 8 or 9
11	exp exercise therapy/
12	exp exercise/
13	exp exercise test/
14	exp exercise movement techniques/
15	exp exercise tolerance/
16	exp physical therapy speciality/
17	(exercise* or physical exercise* or exercise therap* or exercise test* or mobil* or walk* or resistance training or physical activit* or exercise-based intervention* or exercise program* or exercise rehabilitation or physical rehabilitation or ambulat* or gait training).mp
18	(pulmonary rehabil* or rehabil*).mp
19	(frailty or activit* of daily living or ADL or muscle weakness or muscle strength or functional exercise capacity or functional mobility).mp
20	11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19
21	6 and 10 and 20

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Primary Subject Heading:	Rehabilitation medicine
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Authors:

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ABSTRACT

Introduction: Pulmonary rehabilitation (PR) following an acute exacerbation of COPD (AECOPD) reduces the risk of hospital admissions, and improves physical function and health-related quality of life. However, the safety and efficacy of in-hospital PR during the most acute phase of an AECOPD is not well-established. This paper describes the protocol for a systematic review with meta-analysis to determine the safety and efficacy of inpatient acute care pulmonary rehabilitation during the hospitalization phase.

Methods and analysis: Medical literature databases and registries MEDLINE, EMBASE, PEDro, CINAHL, CADTH, CENTRAL, AMED, WHO trials portal, and ClinicalTrials.gov will be searched for articles from inception to June 2021 using a prespecified search strategy. We will identify randomized controlled trials that have a comparison of in-hospital PR with usual care. PR programs had to commence during the hospitalization and include a minimum of two sessions. Title and abstract followed by full-text screening will be conducted independently by two reviewers. A meta-analysis will be performed if there is sufficient homogeneity across selected studies or groups of studies. The Population, Intervention, Comparator, Outcomes, and Study characteristics framework will be used to standardize the data collection process. The quality of the cumulative evidence will be assessed using the Grading of Recommendations, Assessment, Development and Evaluations framework.

Ethics and dissemination: AECOPD results in physical limitations which are amenable to PR. This review will assess the safety and efficacy of in-hospital PR for AECOPD. The results will be presented in a peer-reviewed publication and at research conferences. Ethical review is not required for this study.

Prospero registration number: 198877

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Keywords: Respiratory infections, Chronic airways disease, Rehabilitation Medicine

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This review benefits from a comprehensive search strategy across bibliographic databases, trial registries, scientific conferences, and a grey literature database, ensuring the inclusion of as many relevant articles as possible.

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INTRODUCTION

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A 2016 Cochrane systematic review by Puhan et al,[5] examined the effects of "early" PR (i.e. during or shortly after an AECOPD) on future hospital admissions and other patient-related outcomes such as mortality, health-related QoL, and exercise capacity in patients with COPD. However, this review included studies in which the PR intervention began in the hospital and was continued after discharge. The outcome measures assessed the effect of rehabilitation continued after discharge. It is possible that much of the benefit of PR during an AECOPD is seen during this post-discharge period. What is not clear is if a PR intervention delivered during the hospital admission is safe and results in changes in health status and physical function prior to discharge.

The Cochrane review authors concluded that early PR “improves quality of life and exercise capacity and is a safe intervention for patients with COPD after they have experienced an exacerbation”,[5].

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Despite this, however, the 2017 American Thoracic Society / European Respiratory Society guideline for the management of AECOPD provided a conditional recommendation against the initiation of PR during hospitalization, [6]. This caution was based on their own analysis that found that although PR initiated during hospitalization improved exercise capacity, it also increased mortality. The increased mortality finding was strongly influenced by one study,[7] which reported an increase in all-cause mortality in the intervention group. The excess mortality rate in the intervention group was detected at the 12-month post-discharge assessment, but did not occur during hospitalization, as there was no difference in hospital length of stay or mortality between the groups during the hospital period. Nevertheless, the question remains regarding the safety of rehabilitation specifically during this acute care, inpatient phase, when patients are most ill with an AECOPD,[6]. Clinicians who are considering implementing in-hospital PR for patients soon after admission must be confident when referring their patients to these programs. Therefore, the purpose of this systematic review is to evaluate the safety and efficacy of in-hospital PR for patients admitted to hospital for an AECOPD.

METHODS AND ANALYSIS

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This review will utilize both keyword search terms as well as MeSH (Medical Subject Headings) terms in both MEDLINE and CENTRAL. In Embase we will utilize Emtree terms and in CINAHL we will utilize CINAHL headings. A key "term" search strategy will be employed for the PED*ro*, ClinicalTrials.gov, and WHO trials portal database. The proposed search strategies for MEDLINE is listed in Appendix 1 and will be adapted for use in the other databases. An academic librarian is a member of the study team and is involved in all aspects of developing the search strategies. The reference lists of all primary studies and review articles will be checked for additional references.

Selection of studies

The selection of studies for inclusion will be conducted in two stages. Two reviewers will independently screen the titles and abstracts of all potential studies identified in the search. The studies will be coded as "Yes" or "No". The full-text publications of included citations will then be screened by two reviewers according to our inclusion and exclusion criteria. Studies not in English will be translated using Google Translate. Any discrepancies at any stage will be resolved by discussion and/or consultation with a third reviewer.

Data collection process

Two reviewers will independently collect information from all selected studies using a pre-specified data extraction form. The form will be piloted on the first six selected studies and refined, as necessary. Any discrepancies will be resolved by discussion and/or consultation with a third reviewer. We will contact authors of eligible studies for any missing data. Using the Population, Intervention, Comparator, Outcomes, Study Characteristics framework (PICOS) we will collect information on study

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design, interventions, participants, outcome measures and adverse events to systematize our data extraction (Table 1).

Table 1. PICOS data elements

Population	• Sample size • Age • Gender • Severity of condition • Baseline lung function • Withdrawals
Intervention	• Intervention • Timing • Number of sessions • Frequency of intervention • Supervision provided
Comparators	• Identification and definition of unexposed individuals
Outcomes	• Pre-intervention and post-intervention values of primary and secondary outcomes • Time points when data was collected • Adverse events related to the intervention, including but not limited to: reports of falls, increased length of stay, or responses to exercise (dizziness, chest

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	pain, arrhythmia, hypo/hypertension, increased pain)
Study characteristics	• Funding for studies and notable conflicts of interest of trial authors • Study design • Date of study

Data Analysis and Risk of Bias

Data will be synthesized by calculating mean differences and pooled odds ratios using random-effects models in Review Manager 10,[9]. Heterogeneity will be assessed using forest plots, the Q statistic and the I^2 statistic. Heterogeneity will be examined using subgroup analysis and by analyzing items from the quality assessment that indicate high risk versus low risk of bias (for example, blinding during data collection procedures). High heterogeneity will be considered present if $p<0.1$ for the Q statistic or if I^2 is >50% ,[10]. If we find the included studies are sufficiently homogenous in terms of design, study population and outcomes, we will conduct a meta-analysis to calculate a pooled effect estimate. If prohibited by heterogeneity, a narrative synthesis will be conducted instead.

Risk of bias assessment will be conducted independently by two reviewers on all studies using the criteria outlined in the Revised Cochrane Risk of Bias Tool for Randomized Trials,[11]. Any disagreements will be resolved by discussion and/or consultation with a third reviewer. To assess differences in health-related QoL, we will determine whether estimates and 95% confidence limits between study groups exceed the minimal important difference (MID) for the Chronic Respiratory Questionnaire (± 0.5 points),[12], the St George's Respiratory Questionnaire (± 4 points),[13], the COPD Assessment Test (2 points),[14], the 6-minute walk test (≥ 30 meters),[15] and the incremental

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shuttle walk test (≥ 47.5 meters),[16]. The unit of analysis is the participant. Patients who receive the usual care will be used as the control group in this study.

In order to account for selective reporting within studies, we will assess all available protocols for the included studies using the trial registries ClinicalTrials.gov and the WHO trials portal. Publication bias across studies will be assessed using funnel plots. The quality of the cumulative evidence for each outcome will be assessed using the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) framework which includes the risk of bias, consistency, directness, precision and publication bias.

Patient and Public Involvement

There was no patient involved in the development of this protocol.

ETHICS AND DISSEMINATION

Ethical review is not required as this study is a systematic review. It is our intention to submit the results of our review for peer-reviewed publication and to present our findings at scientific conferences.

DISCUSSION

Pulmonary rehabilitation plays an important role in the management of AECOPD. A large body of evidence shows that pulmonary rehabilitation in patients with AECOPD helps to improve QoL, symptom management, and disease prognosis,[5]; however, to our knowledge, there has been no review to assess the acute in-hospital use of pulmonary rehabilitation for AECOPD. This review will determine the safety and efficacy of initiating physical rehabilitation, in patients with AECOPD, during their hospitalization and will provide suggestions for future guidelines and recommendations.

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Contributors

KZ, JG, and PC drafted the protocol and developed the study design and the data extraction form. KZ, JG, AK, JC, AE, SC, HD, HP, and PC contributed to the development of the search strategies. All authors read and approved the final version of the manuscript for submission.

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Competing Interests.

None declared.

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Appendix 1. Database search strategies

Database: MEDLINE® ALL <Inception to June, 2020>

Platform: Ovid SP

Search Line	Search Item
1	exp pulmonary disease, chronic obstructive/
2	exp lung diseases, obstructive/
3	(COPD or COAD or AECOPD or COBD or AECB or (acute exacerbation adj2 (COPD or chronic obstructive pulmonary disease))).mp
4	((chronic or obstruct*) adj3 (airflow or airway* or bronch* or lung* or pulmonary or respirat*)).mp
5	(emphysema or lung hyperinflation or hyperlucent lung or enlarged alveolar spaces).mp
6	1 or 2 or 3 or 4 or 5
7	exp hospitalization/
8	exp inpatients/
9	(((transitional care or step down or subacute or acute care or medical acute or critical care) adj3 (center* or centre* or unit* or floor* or ward* or clinic*)) or (hospital* or intensive care or ICU or inpatient* or in-patient* or patient* or emergency department*)).mp
10	7 or 8 or 9
11	exp exercise therapy/
12	exp exercise/
13	exp exercise test/
14	exp exercise movement techniques/
15	exp exercise tolerance/
16	exp physical therapy speciality/
17	(exercise* or physical exercise* or exercise therap* or exercise test* or mobil* or walk* or resistance training or physical activit* or exercise-based intervention* or exercise program* or exercise rehabilitation or physical rehabilitation or ambulat* or gait training).mp
18	(pulmonary rehabil* or rehabil*).mp
19	(frailty or activit* of daily living or ADL or muscle weakness or muscle strength or functional exercise capacity or functional mobility).mp
20	11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19
21	6 and 10 and 20

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Manuscript Location
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Page 1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Page 2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Page 1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Page 10
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Page 10
Sponsor	5b	Provide name for the review funder and/or sponsor	n/a
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	n/a
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Page 4
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Page 8
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Page 5-6
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Page 6
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Page 13

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Page 7
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	Page 7
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	Page 7
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Page 8
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Page 8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Page 9
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	Page 9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Page 9
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	Page 9
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n/a
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Page 10

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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